

Md. Ragib Hasan

Dhaka, Bangladesh | +8801903156620 | ragibhasansec@gmail.com
[GitHub](#) | [LinkedIn](#) | [Portfolio](#) | [Google Scholar](#)

Education

Bachelor of Science in Computer Science and Engineering

Sylhet Engineering College, Sylhet, Bangladesh

Affiliated with Shahjalal University of Science and Technology (SUST)

Aug 2026

CGPA: 3.97/4.00

Research Interests

- Computer vision and deep learning for real-world visual understanding, from everyday scenes to traffic and transportation settings
- Object-centric recognition, OCR, and document intelligence
- Multimodal learning
- Efficient and robust systems for challenging, real-world environments
- Applications in healthcare and low-resource language and script recognition

Academic Experience

Adjunct Lecturer

Department of Computer Science and Engineering, Sylhet Engineering College

Present

Research Experience

Research Assistant

Department of Computer Science and Engineering, Sylhet Engineering College

Present

Working under Md. Abu Naser Mojumder, Associate Professor and Head

- Current research focuses on real-world computer vision problems, including architectural enhancements to Ultralytics /YOLO-based detectors, OCR and handwritten script recognition, and continued work extending traffic and vehicle-detection research beyond the undergraduate thesis.

Undergraduate Thesis

From Traffic Scenes to Registration Strings: A Field Study of Vehicle Detection and Bangla License Plate Recognition in Bangladesh

Md. Ragib Hasan, Naimur Rahman, Supto Saha

First Author | Supervisor: Md. Abu Naser Mojumder, Associate Professor and Head, Department of CSE, Sylhet Engineering College

- Collected and curated real-world traffic video data from five locations across Dhaka and Sylhet, and evaluated 13 detection and 9 OCR configurations for vehicle detection and Bangla license-plate recognition under unconstrained conditions; YOLO11n-SOEN achieved 0.6142 mAP@0.50:0.95, while BPNet achieved 82.36% test accuracy and 67.07% unseen-string accuracy.
- Developed an end-to-end video inference pipeline integrating vehicle tracking, plate-to-vehicle association, quality-aware OCR selection, and temporal result aggregation.

Publications

- **ChakmaCNN: A Pronunciation-Aware CNN for Handwritten Chakma Script Recognition**
Md. Ragib Hasan, Kakoli Rani Ray Koli, Naimur Rahman
ICCIT 2025 [IEEE Xplore](#)
- **DateFNet: From Pixels to Plates: Attention Based Multi-Stream CNN for Date Fruit Classification**
Arif Rahman Samad, Munshaibur Rahman Mahin, Md. Ragib Hasan, Joydrath Roy Joy, Fahmiduzzaman Fahmiduzzaman, Gourab Roy, Md. Abu Naser Mojumder
ICCIT 2025 [IEEE Xplore](#)
- **Towards Secure Digital Communication: Deep Learning-Based Automated Classification of Malicious Bangla Messages**
Sandwip Dash Talukder, Mahmudul Hasan Badhan, Abul Kasim Mohammad Abu Musa, Md. Saidul Islam, Md. Ragib Hasan, Islam Md. Saiful
QPAIN 2026 [IEEE Xplore](#)

Manuscript Under Review

Performance and Generalization Analysis of CNN and Hybrid Deep Learning Models for Handwritten Medicine-Name Recognition in Bangladeshi Prescriptions

Md. Ragib Hasan, Supto Saha, Munshaibur Rahman Mahin, Moin Uddin Sayem, Pushpita Sikder, Dipta Das Gupta, Md. Abu Naser Mojumder

Intelligence-Based Medicine, Elsevier

Submitted Jul 2026

- Evaluates seven deep learning architectures on a 22,190-image, 317-class Bangladeshi handwritten medicine-name dataset, including cross-dataset evaluation to assess generalization.

Research in Progress

Lightweight CNN Models for Handwritten Hangul Alphabet Classification

- Investigating lightweight handwritten-script recognition with CNN-based models for efficient recognition in low-resource settings.

Submitted Conference Papers

Submitted to ICCIT 2026; currently under conference evaluation.

- **RGCF-YOLO11: Region-Gated Context Fusion for Tiny PCB Defect Detection on Periodic Textures**
Md. Ragib Hasan, Pushpita Sikder, Sk Md Rafiul Karim, Gourab Roy, Nafiz Nahid, Md. Abu Naser Mojumder
- **Content-Adaptive Multi-Scale Tokenization for Lightweight ViTs in Remote Sensing**
Md. Ragib Hasan, Yeasin Hasan, Md. Nayeem Hossain, Abdullah Bin Hasan, Adnan Hosen
- **BongoBarta: A GRU-Attention Model with FastText Embeddings for Bengali News Headline Classification**
Supto Saha, Md. Ragib Hasan, Munshaibur Rahman Mahin, Sizen Mahmud, Nafiz Nahid
- **YGP-Net: A Lightweight P6-Enhanced YOLO11 Network for Efficient Obstacle Detection**
Pushpita Sikder, Md. Ragib Hasan, Sk Md Rafiul Karim, Gourab Roy, Nafiz Nahid, Md. Abu Naser Mojumder

Additional papers have also been submitted as a co-author.

Technical Skills

Programming: C++, Python, Java, JavaScript, TypeScript
ML/DL: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas
Computer Vision: OpenCV, Ultralytics/YOLO, OCR, Image Processing
Web: Next.js, React, Node.js, Django
Tools: Git, GitHub, LaTeX

Selected Projects

- BloBax** AI Healthcare Platform
• Independently developed a Django-based AI healthcare platform integrating health prediction, handwritten medicine-name recognition, and AI-assisted emergency guidance; integrated a medicine-name recognition model trained as part of my research.
• **Technologies:** Python, Django, Gemini API, Machine Learning, MySQL [GitHub](#)
- BioEnclave** Full-Stack Web Application
• Independently designed and developed a production web application as a paid client project.
• **Technologies:** Next.js, TypeScript, Tailwind CSS, Cloudflare, Vercel [Live Website](#)
- BizTrade Venture** Full-Stack Web Application
• Independently developed a product and administration management application as a paid client project for a Malaysian wholesale company.
• **Technologies:** Next.js, React, Node.js, Tailwind CSS
- In addition to the projects above, I have independently completed several other freelance and personal projects, and continue to work on new projects on an ongoing basis.*

Leadership & Academic Activities

- Vice President**, SEC CSE Research Club Present
General Secretary, SEC CSE Society Oct 2025 – Aug 2026 | [Certificate](#)
Organizing Secretary, SEC Programming Club Oct 2025 – Aug 2026
Participant, ICCIT 2025 | [Certificate](#)
Participant, SEC CSE Fest 2k22 – Hackathon Event | [Certificate](#)
- I have also participated in a number of other onsite programming contests, idea presentations, and related departmental activities.*

Additional Achievements

Solved 1200+ problems across Codeforces, LightOJ, UVa and other online judges; achieved Pupil rating on Codeforces.
[Codeforces Profile](#)

Certification

Supervised Machine Learning: Regression and Classification Jun 9, 2025
DeepLearning.AI / Stanford Online, Coursera | [Certificate](#)

References

- Md. Abu Naser Mojumder**
Associate Professor and Head
Department of Computer Science and Engineering
Sylhet Engineering College
abu.naser84@gmail.com | +8801712291794
- Mohammad Shahidur Rahman, PhD**
Professor
Department of Computer Science and Engineering
Shahjalal University of Science and Technology
rahmanms@sust.edu